

Exmpo I

$$1) N = q \times d + r$$

$$\begin{array}{r} -27 \ 18 -13 \\ \underline{-27} \end{array}$$

$$8 + \frac{13}{27}$$

$$\begin{array}{r} 27 \\ \times 8 \\ \hline 216 \end{array} \quad \begin{array}{r} 5 \\ 229 \mid \underline{27} \\ 14 \quad 9 \end{array}$$

$$\begin{array}{r} 27 \\ \times 7 \\ \hline 189 \end{array}$$

$$-229 = (-27) \times 9 + 14$$

(F)

$$2) \checkmark$$

$$3) \checkmark$$

$$4) F$$

$$5) F$$

$$6) 194 \times (-22) \equiv_5 2 ?$$

$$194 \equiv_5 4 \quad ; \quad -22 \equiv_5 3$$

$$194 \times (-22) \equiv_5 4 \times 3 \equiv_5 2 \equiv_5 -3$$

(F)

$$7) F$$

$$8)$$

(V)

$$9) F$$

$$10) \checkmark$$

Cmho IV

$$1) \quad 2184 = 996 \times 2 + 192$$

$$996 = 192 \times 5 + 36$$

$$192 = 36 \times 5 + 12$$

$$36 = 12 \times 3 + 0$$

$$\text{m.d.c.}(2184, 996) = 12$$

$$12 = 192 - 36 \times 5$$

$$= 192 - (996 - 192 \times 5) \times 5$$

$$= 192 \times (26) - (5) \times 996$$

$$= (2184 - 996 \times 2) \times 26 - 5 \times 996$$

$$= 2184 \times (26) + 996 \times (-57)$$

$$2) \quad \text{m.d.c.}(102, 258) = 6$$

$$258 = 2 \times 102 + 54$$

$$102 = 1 \times 54 + 48$$

$$54 = 1 \times 48 + 6$$

$$48 = 8 \times 6 + 0$$

$$x = 30 - 43k, k \in \mathbb{Z}$$

$$x = \{116, 159, 202, 245, 288, \dots\}$$

$$204$$

$$\begin{array}{r} 60 \\ 54 \\ 48 \end{array} \quad \begin{array}{l} 1 \\ 1 \\ 1 \end{array}$$

$$\begin{array}{r} 258 \\ 18 \end{array} \quad \begin{array}{l} 16 \\ 43 \end{array}$$

$$3) 2^{136} \equiv_{13} (2^6)^{23} \equiv_{13} (-1)^{23} \equiv_{13} -1 \equiv_{13} 12$$

$$2^6 \equiv_{13} -1$$

$$\begin{array}{r} 13 \overline{) 6} \\ 18 \quad 23 \end{array}$$

$$4) 4 + 3 + 5 + n + y = 9K, \quad K \in \mathbb{N}_0$$

$$12 + n + y = 9K$$

$$n=1 \leftrightarrow y=5$$

$$n=2 \leftrightarrow y=4$$

$$\checkmark n=3 \leftrightarrow y=3$$

$$n=4 \leftrightarrow y=2$$

$$n=5 \leftrightarrow y=1$$

$$n=6 \leftrightarrow y=0, y=9$$

$$\checkmark n=7 \leftrightarrow y=8$$

$$n=8 \leftrightarrow y=7$$

$$n=9 \leftrightarrow y=6$$

40
44
46
52
56
60

$$n=3 \vee n=7$$

$$S = 343353; 438574$$

Exercício 10

$$1) \text{ m.d.c.}(22, 27) = 1$$

$$27 = 1 \times 22 + 5$$

$$22 = 4 \times 5 + 2$$

$$5 = 2 \times 2 + 1$$

$$2 = 2 \times 1 + 0$$

Como $1 \mid 12$, a equação tem solução

$$\begin{aligned} 1 &= 5 - 2 \times 2 = 5 - 2(22 - 4 \times 5) \\ &= 3 \times 5 - 2 \times 22 \\ &= (27 - 22) \times 3 - 2 \times 22 \\ &= 22 \times (-1) + 27 \times 3 \end{aligned}$$

$$12 = 22 \times (-13) + 27 \times (10)$$

$$x = -13 + 27k$$

$$y = 10 - 22k, \quad k \in \mathbb{Z}$$

$$2) \quad 84x \equiv_{228} 204 \Leftrightarrow 84x - 228y = 204$$

$$\text{m.d.c.}(84, 228) = 12$$

$$228 = 2 \times 84 + 60$$

$$84 = 1 \times 60 + 24$$

$$60 = 2 \times 24 + 12$$

$$24 = 2 \times 12 + 0$$

Como $12 \mid 204$, a equação tem solução

$$12 = 60 - 2 \times 24$$

$$= (228 - 2 \times 84) - 2(84 - 60)$$

$$= 228 - 4 \times 84 + 60 \times 2$$

$$= 228 - 4 \times 84 + (228 - 2 \times 84) \times 2$$

$$= 84 \times (-8) - 228(-3)$$

$$208 = 84 \times (-136) - 228(-51)$$

$$n = -136 - 19K, K \in \mathbb{Z}$$

$$\text{For } K = -4$$

$$n = -60$$

$$\text{For } K = -5$$

$$n = -41$$

$$\text{For } K = -6$$

$$n = -22$$

$$\text{For } K = -7$$

$$n = -3$$

$$\text{For } K = -8$$

$$n = 16$$

$$\text{For } K = -9$$

$$n = 35$$

$$\text{For } K = -10$$

$$n = 54$$

$$CS = \{-41, -22, -3, 16, 35\}$$

$$3) \begin{cases} n \equiv_5 3 \\ n \equiv_8 5 \\ n \equiv_{11} 1 \end{cases} \Leftrightarrow \begin{cases} n = 3 + 5K \\ 3 + 5K \equiv_8 5 \\ 3 + 5K \equiv_{11} 1 \end{cases} \Leftrightarrow \begin{cases} 5K \equiv_8 2 \\ 5K \equiv_{11} 9 \end{cases}$$

$$\Leftrightarrow \begin{cases} 15K \equiv_8 6 \\ 10K \equiv_{11} 18 \end{cases} \Leftrightarrow \begin{cases} -K \equiv_8 -10 \\ -K \equiv_{11} 7 \end{cases} \Leftrightarrow \begin{cases} K \equiv_8 10 \\ K \equiv_{11} -7 \end{cases}$$

$$\Leftrightarrow \begin{cases} K = 10 + 8u \\ 10 + 8u \equiv_{11} 4 \end{cases} \Leftrightarrow \begin{cases} 8u \equiv_{11} 5 \end{cases} \Leftrightarrow \begin{cases} 32u \equiv_{11} 20 \end{cases}$$

$$\Leftrightarrow \begin{cases} -u \equiv_{11} 9 \end{cases} \Leftrightarrow \begin{cases} u \equiv_{11} 2 \end{cases} \Leftrightarrow \begin{cases} u = 2 + 11t \end{cases}$$

$$\begin{aligned} n &= 3 + 5(10 + 8(2 + 11t)) \\ &= 3 + 5(10 + 16 + 88t) \\ &= 133 + 440t, \quad t \in \mathbb{Z} \end{aligned}$$

A number only negative if for $t = -1$

$$\text{once } n = -367$$